Unit-1

1. Explain why Python is preferred for Data Science. Mention at least two of its features that support data analysis tasks.
2. How do you launch IPython from the command line?
3. Name and describe the purpose of any four fundamental Python libraries commonly used in Data Science.
4. How can you access help and documentation in IPython? Mention at least two ways of obtaining function or object documentation.
5. What are the benefits of using IPython over the regular Python shell?
6. Define Data Science. What are its key components and applications in modern industries?
7. How does IPython differ from the standard Python shell? Mention at least two advanced features it offers.
8. How do you launch a Jupyter Notebook? Briefly explain its use in data analysis and visualization.
9. Why is Python considered a powerful tool for Data Science? Discuss its advantages with respect to readability, community support, and integration with data science libraries and tools.
10. What is an IDE? Compare and contrast at least two popular Python IDEs used in Data Science in terms of features, usability, and productivity.
11. Discuss the importance of keyboard shortcuts in improving productivity within the IPython Shell. List and explain the use of at least six shortcuts commonly used by data scientists.
12. What are magic commands in IPython? Differentiate between line and cell magics. Provide examples of at least four magic commands and describe their use in simplifying complex tasks.

Unit-2

1. What is NumPy, and why is it important in Python for numerical computations?
2. How do you create a NumPy array? Mention any two attributes of a NumPy array and explain their significance.
3. What are the two main data structures in Pandas? Briefly explain each with an example.

4. What is a scatter plot? Write a short example using plt.scatter() to visualize a dataset.
5. How does NumPy sort arrays? Differentiate between np.sort() and ndarray.sort() with one example.
6. Explain how aggregation functions like min(), max(), mean() and Aggregating Along an Axis are used in NumPy. Provide one code example.
7. Write a Python program using Matplotlib to create a line plot for a dataset. Add labels, a title, and a legend. Explain each component used in the plot.
8. What functions does Pandas provide to detect and handle missing data? Mention two methods for dealing with missing values.
9. Explain the structure and key attributes of NumPy arrays. How can arrays be created using functions like np.array(), np.zeros(), and np.arange()? Provide examples and explain each.
10. Describe Series and DataFrame objects in Pandas. What are their key characteristics? How are they created? Compare and contrast Series and DataFrame with relevant code examples
11. How do you save a plot using savefig() in Matplotlib? Discuss the parameters such as filename, dpi, and format. Provide examples of saving a plot as PNG and PDF.
12. What are the key components of a plot in Matplotlib? Discuss general tips for making effective visualizations such as using titles, labels, legends, and styles. Provide examples.

Unit-3

1. Mention four key objectives of descriptive statistics.
2. What is data preparation in descriptive statistics? Why is it needed?
3. Why is statistical inference important in data analysis?
4. What is a p-value in hypothesis testing?
5. Explain the measures of dispersion (range, variance, standard deviation) and their role in descriptive statistics.
6. Describe the steps involved in data cleaning, integration, and transformation.
7. Define and explain standard error. How does it change with sample size? Illustrate with an example.

8. Differentiate between population variance, sample variance, and standard error with examples.
9. What is data preparation? Explain in detail the steps involved in that.
10. Describe the importance of data cleaning and transformation in preparing data for descriptive statistics.
11. What is Exploratory Data Analysis (EDA)? Explain its objectives and importance in data science.
12. Explain in detail the steps of hypothesis testing with an example.

Unit-4

1. What are the key components of supervised learning?
2. Why is generalization important in machine learning?
3. Define Random Forest in machine learning?
4. Mention one application of logistic regression in real life?
5. Discuss why splitting the dataset into training, validation, and test sets is necessary for supervised learning?
6. Discuss two major types of regression models used in data science?
7. What is logistic regression? Explain its working with the logistic (sigmoid) function and graph?
8. How can learning curves help in identifying overfitting and underfitting? Illustrate with examples?
9. What is Random Forest? Explain its construction using decision trees with an example?
10. Explain logistic regression. Why is it used for classification instead of linear regression?
11. Differentiate between linear and polynomial regression. Why is polynomial regression used for non-linear data?

Unit-5

1. Differentiate between supervised and unsupervised learning?
2. Write any two commonly used distance measures in clustering?
3. What is K-means clustering.
4. Name two metrics used to evaluate clustering quality.
5. Explain the Role of Similarity and Distance in Clustering.
6. Explain the properties of a good clustering with examples.

7. Discuss internal and external evaluation metrics used in clustering.
8. Explain the working of the K-means clustering algorithm with a diagram.
9. Compare supervised, unsupervised, and semi-supervised learning with examples.
10. Explain how the choice of distance metric can affect clustering results. Give examples with diagrams.
11. Why are similarity and distance measures crucial in clustering Illustrate with real-world data analysis examples.
12. Explain the spectral clustering algorithm in detail with steps and an example.